

PROTEINS

- Proteins are essential for maintaining and sustaining life.
- Proteins are large molecules composed of chains of amino acids, which are simple subunits made of carbon, oxygen, hydrogen, nitrogen, sulphur and some also contain phosphorus.
- The word protein comes from a Greek word meaning “of primary importance,” which is an apt description of these macronutrients as they are also known as the “workhorses” of life.
- Proteins provide structure to bones, muscles, and skin, and they play a role in conducting most of the chemical reactions occurring in the body.



- Proteins are the indispensable constituents of the diet. Proteins are made up of amino acids. Amino acids are needed for replacement and growth of the body parts.
- Proteins are chains of amino acids held together by peptide bonds, only about 20 different amino acids are used but the number of ways in which they can be arranged is almost infinite.
- It is the specific and unique sequence of these units which gives each protein its characteristic structural and enzymatic properties.
 - Example: Egg white protein, is very sensitive. It denatures on heating, dissolves easily in water and is quite reactive, while the other one is keratin of nails and hoofs, is wholly insoluble, tough and chemically inert and resistant.

- Food sources of proteins include meats, dairy products, seafood, and a variety of plant-based foods, like beans, nuts, and seeds.
- Proteins can also provide energy, though this is a relatively minor function, as carbohydrates and fat are preferred energy sources.

- Classification of Proteins

Classification based on chemical nature

- i) Simple proteins (albumin, globulin, glutelin, prolamins, fibrous proteins, histones)
- ii) Conjugated proteins (glycoproteins, lipoproteins, nucleoproteins)
- iii) Derived proteins (metaproteins, coagulated proteins, proteoses, peptones, peptides)

Classification based on chemical properties

- i) Mono amino mono carboxylic acids
- ii) Mono amino dicarboxylic acids
- iii) Diamino mono carboxylic acids
- iv) Sulphur containing amino acids
- v) Aromatic and heterocyclic amino acids

Classification based on amino acid content

- i) Essential (indispensable)
- ii) Conditionally essential
- iii) Non essential (dispensable)

Amino acids are classified as essential and non-essential amino acids

- Essential amino acids cannot be synthesized by the body and must be taken through foods
- Non-essential amino acids can be synthesized by the body.

a. Complete protein:

- Complete proteins are proteins that contain essential amino acids
- Plant-based foods do not require complete protein.
- Examples are animal foods such as meat, fish, milk, and eggs

b. Incomplete protein:

- Incomplete proteins are proteins that contain all the essential and non-essential amino acids
- Examples are vegetable foods such as cereals, legumes, and peanuts

- Sources of Protein

- The important sources of proteins in the diets are cereals and legumes but the overall proportion of amino acids in any single vegetable food is less than needed by man. These proteins are therefore known to have low biological value.
- Most animal proteins like meat, fish, eggs and milk are important sources of proteins and have high biological value. Oilseeds, oilseed meals and soy are also rich sources of proteins.

- Functions of proteins

- Bodybuilding functions of proteins

- The primary functions of proteins are tissue growth and maintenance.
- Protein contains amino acids- the building blocks- that our bodies use to build and maintain muscles, bone, skin, blood and other organs. Thus, proteins play an important role in growth and body-building.

- Protein as an energy source:

- If diet does not furnish enough calories from carbohydrates and fats, proteins contribute to the body's energy need by catabolizing. One gram protein yields 4 Kcal.
- But it is important to understand that this is not the major function of proteins. This only happens when the diet does not supply enough energy-giving nutrients.

- Proteins as enzymes

- From conception to death, living cells use oxygen and metabolize fuel. Cells synthesize new products, degrade others, and generally are in a state of metabolic flux. For these processes to occur, catalysts are needed. These catalysts called 'enzymes' are proteins.
- Enzymes are the largest and the most specialized class of proteins. Each enzyme is unique and catalyzes a specific kind of reaction. In the cell, enzymes are found in cellular compartments (cytoplasm, nucleus, mitochondria, etc.), as well as the membranes within and around the cell wall.

- Proteins as carriers

- A large variety of compounds are carried in the blood between tissues and organs of the body. Some of the compounds require specific protein for their transport.

- Proteins as regulators of water balance:

- As substrates and solutes are transferred or exchanged across membranes, water has a tendency to flow to maintain equal osmotic pressure on each side of the membrane.
- If osmotic pressure is not maintained, the individual cells either shrink from lack of internal water or burst from too much. Albumin plays a significant role in maintaining the osmotic pressure because of its size and abundance.

- The protein hormone particularly the Anti-Diuretic Hormone (ADH) plays a role in controlling water balance. The effect of protein is on the distribution of water amongst the various body compartments than on the total body water.

- Proteins as biological buffers

- Proteins have the ability to accept or donate hydrogen ions and by doing so they serve as biological buffers. In blood, there are three important buffering systems - plasma proteins, hemoglobin and carbonic acid bicarbonate.

- Proteins as structural elements and structural units
 - Proteins function as enzymes, carriers, binding or receptor sites or entry ports for a large variety of materials binding, entering or leaving the cell. Thus proteins serve as the structural and functional units of the cell membrane.
 - The most important structural function of protein is related skin and connective tissue. The skin is composed of epithelial tissue which covers not only the exterior of the body but also lines the gastrointestinal tract, respiratory tract and the urinary tract.
 - One of the major protein found in the skin is 'melanin'. Collagen and elastin are the two distinct types of proteins in the connective tissue. Muscle is composed of 20% protein, 75% water and 5% inorganic material, glycogen and other organic compounds. The major proteins in muscle are myosin - a large globular protein, and actin - a smaller globular protein.

- Proteins as lubricants

- The mucus of the respiratory tract, oral cavity, vaginal tract and the rectal cavity reduces the irritation which might be caused by materials moving through these passages. This mucus is a mucoprotein.
- Proteins as lubricants also surround the joints and facilitate their movement.

- Proteins in the immune system:

- Proteins such as γ -globulin serve to protect the body against foreign cells. The immunoglobulins are produced by lymphocytes.

- Daily requirements

- The ICMR recommends 1gm of protein/ Kg of body weight for adults
- The amount of protein should be increased for children, pregnant and lactating mothers by 1.5- 2 g/ kg body weight.

Protein digestion in the mouth

- Unless you are eating it raw, the first step in digesting an egg (or any other solid food) is chewing.
- The teeth begin the mechanical breakdown of large egg pieces into smaller pieces that can be swallowed.
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- The salivary glands secrete saliva to aid swallowing and the passage of the partially mashed egg through the esophagus.

Protein digestion in the stomach

- The mashed egg pieces enter the stomach from the esophagus
- Both mechanical and chemical digestion take place in the stomach. The stomach releases gastric juices containing hydrochloric acid and the enzyme, pepsin, which initiate the chemical digestion of protein.
- Muscular contractions, called peristalsis, also aid in digestion. The powerful stomach contractions churn the partially digested protein into a more uniform mixture, which is called chyme.

- Because of the hydrochloric acid in the stomach, it has a very low pH of 1.5-3.5. The acidity of the stomach causes food proteins to denature, unfolding their three-dimensional structure to reveal just the polypeptide chain. This is the first step of chemical digestion of proteins.

- Once proteins are denatured in the stomach, the peptide bonds linking amino acids together are more accessible for enzymatic digestion.
- That process is started by Pepsin, an enzyme that is secreted by the cells that line the stomach and is activated by hydrochloric acid.
- Pepsin begins breaking peptide bonds, creating shorter polypeptides

Protein digestion and absorption in the small intestine

- The chyme leaves the stomach and enters the small intestine, where the majority of protein digestion occurs.
- The pancreas secretes digestive juices into the small intestine, and these contain more enzymes to further break down polypeptides.
- The two major pancreatic enzymes that digest proteins in the small intestine are Chymotrypsin and trypsin. Trypsin activates other protein-digesting enzymes called proteases, and together, these enzymes break proteins down to tripeptides, dipeptides, and individual amino acids.

- Tripeptides, dipeptides, and single amino acids enter the enterocytes of the small intestine using active transport systems, which require ATP.
- Once inside, the tripeptides and dipeptides are all broken down to single amino acids, which are absorbed into the bloodstream.
- Proteins that aren't fully digested in the small intestine pass into the large intestine and are eventually excreted in the feces.

- In order to use amino acids to make ATP, the excess nitrogen first has to be removed in a process called deamination, which occurs in the liver and kidneys.
- The nitrogen is initially released as ammonia, and because ammonia is toxic, the liver transforms it into urea.
- Urea is then transported to the kidneys and excreted in the urine. Urea is a molecule that contains two nitrogens and is highly soluble in water. This makes it ideal for transporting excess nitrogen out of the body.

Summary:

Digestion

- Mouth: No protein digestion occurs here; saliva lubricates food.
- Stomach:
 - HCl denatures proteins (unfolds them) and activates pepsinogen into pepsin
 - Pepsin breaks proteins into smaller polypeptides
- Small Intestine:
 - Pancreatic enzymes (like trypsin, chymotrypsin) continue breaking polypeptides into smaller peptides and amino acids
 - Intestinal brush border enzymes (peptidases) break these down into absorbable units: individual amino acids, dipeptides, and tripeptides

Absorption

- Location:

Primarily the jejunum and ileum of the small intestine.

- Mechanism:

- Amino acids, dipeptides, and tripeptides are transported across the intestinal wall enterocytes
- Dipeptides and tripeptides are further broken down into amino acids inside the intestinal cells by cytosolic peptidases.

- Transport:

Free amino acids and small peptides then exit the intestinal cells into the bloodstream.

- Liver Processing:

Amino acids travel to the liver, which acts as a checkpoint for distribution or further processing, with excess nitrogen converted to urea for excretion